This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Event Structural Valley: A Unified Theoretical and Practical Framework for Event Camera Autofocus

Xijie Xiang[1] Lin Zhu[2*] Wei Zhang[3] Yonghong Tian[1*]
[1]Peking University [2]Beijing Normal University [3]Pengcheng Laboratory
{xiangxj,yhtian}@pku.edu.cn linzhu@bnu.edu.cn zhangwei1213052@126.com

Abstract

Autofocus in dynamic environments remains challenging for conventional frame-based sensors, which often fail under fast motion, low light, or high dynamic range conditions. Event cameras, with microsecond temporal resolution and asynchronous brightness detection, offer a promising alternative. However, typical event-based autofocus methods assume that the sharpest focus corresponds to the maximum event rate. In this paper, we reveal a counterintuitive yet consistent phenomenon: the true focus actually corresponds to a local minimum in the event-rate curve. We theoretically derive this behavior from the physics of event generation and show that as defocus blur increases, the event rate first rises and then declines, forming a dual-peak-valley structure across focal distances. Based on this insight, we propose an ***Event Structural Valley-based Autofocus (ESVA)*** *framework that identifies the valley between two dominant peaks as the true focal position. ESVA integrates structural smoothing, consistency filtering, and a dual-peak constraint to robustly recover the valley under noise and motion disturbances. Extensive experiments on multiple synthetic and real-world datasets demonstrate that ESVA delivers accurate and stable focus estimation, consistently outperforming existing event-based autofocus methods without requiring image reconstruction or supervision.*

1. Introduction

Fast and accurate autofocus [15, 38, 43] is essential for dynamic vision systems such as robotics [23, 42], autonomous driving [1, 41], and microscopic imaging [8, 16, 28]. However, conventional frame-based autofocus algorithms often fail under fast motion, low illumination, or high dynamic range conditions [10, 12, 44]. Event cameras [5, 10, 20], with microsecond temporal resolution, low latency, and high dynamic range, inherently overcome these limitations by asynchronously capturing per-pixel brightness changes. Their proven effectiveness in 3D reconstruction [9, 32, 45], visual odometry and SLAM [24, 31, 35], image reconstruction [33, 40, 46] and motion deblurring [25] demonstrates the potential of event-based sensing in challenging environments. This motivates the exploration of event-based autofocus as a principled and promising solution for achieving fast and reliable focus control.

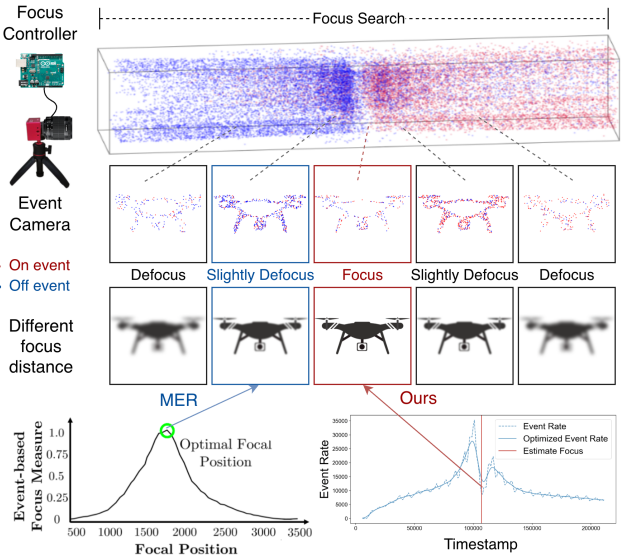


Figure 1. **Event-based Autofocus.** The goal of event-based autofocus is to determine the optimal focus position from asynchronous event streams acquired at different focal depths. During a continuous focus sweep (top), the event camera generates ON/OFF events in response to brightness changes along scene edges. The middle rows show accumulated event maps and corresponding intensity images across focal settings. Bottom left: previous Maximum Event Rate (MER) methods [21, 22] select the focus with maximum event activity, often yielding approximate results at slightly defocused states; bottom right: our structural valley-based approach locates the inter-peak valley to achieve a more accurate focus estimate.

As illustrated in Figure 1, event-based autofocus aims to estimate the focal distance that yields the sharpest scene by relying solely on asynchronous events captured during a continuous focus sweep. During this process, a motor-

*Corresponding author: Yonghong Tian, Lin zhu.

ized lens controller synchronously adjusts the focal plane while the event camera records brightness changes, forming a continuous spatiotemporal event volume.

Representative prior works [21, 22] assume that the sharpest focus corresponds to the *maximum* event rate (bottom left of Figure 1), following the intuition that clear focus produces more events. However, both theoretical analysis and experimental evidence revealed a counterintuitive observation: the true in-focus position corresponds to an *inter-peak minimum* in the event-rate curve. As shown in Figure 1, our approach identifies this structural valley, which consistently aligns with the physically sharpest focus.

This inverse trend arises from the physics of event generation: at perfect focus, edges are spatially compact, so fewer pixels cross the contrast threshold and fewer events are triggered [9, 19]. When slightly defocused, blur broadens the transition zone, activating more pixels and increasing the event rate [9, 20]. As defocus increases further, gradients weaken and the event rate decreases, forming a characteristic dual-peak–valley structure over the focal range. This structure consistently indicates that the sharpest focus lies at the valley between two dominant peaks.

Recent studies [2, 11] reported polarity inversion and symmetry of event activity near focus, providing empirical insights into event behavior around the focal plane. However, these observations were obtained under constrained setups (e.g. microscopic autofocus) and interpreted through heuristic cues such as polarity balance, without uncovering the physical origin of the valley itself. In contrast, we develop a unified analytical model that derives this behavior directly from the event generation process and analytically shows that the inter-peak minimum of the event-rate curve inherently corresponds to the true focus.

Building on this insight, we propose a simple yet robust *Event Structural Valley-based Autofocus (ESVA)* framework. Our method detects two dominant peaks in the event-rate curve and locates the valley between them as the optimal focus estimate. It operates fully in the event domain without any image reconstruction or supervision, and performs consistently across real and synthetic datasets. Moreover, it provides a unified physical and computational foundation for event-based focusing.

Our main contributions are:

- We establish a theoretical model that explains why the event rate exhibits a dual-peak valley structure during focal scanning and analytically show that its valley corresponds to the true focus.
- We propose a structural, valley-based autofocus algorithm integrating smoothing, consistency, and dual-peak modules for robust and accurate focus estimation.
- Extensive experiments demonstrating superior accuracy and generalization over prior event autofocus methods in multiple datasets, without any image-based supervision.

2. Related Work

Event-based Autofocus Methods. Event cameras have recently been adopted for autofocus to enable real-time, low-latency focus estimation under challenging conditions such as high-speed motion and extreme lighting variations. Lin *et al*. [21] first employed the event rate peak as a focus metric combined with a golden-section search strategy, assuming that an in-focus scene trigger more events. However, both our analysis and empirical evidence show that this assumption fails at perfect focus, where edges are spatially compact and thus produce fewer events.

To improve robustness, subsequent works refined event-driven metrics. Bao *et al*. [2] introduced polarity symmetry as a focus cue, observing that well-focused scenes exhibit balanced positive and negative events. Ge *et al*. [11] further identified a symmetric polarity distribution around the focal plane in microscopic imaging. Qu *et al*. [29] extended this idea to a more robust autofocus framework for microscopes, reinforcing the potential of neuromorphic sensing for precise axial focusing. In parallel, Su *et al*. [36] developed an illumination-robust autofocus metric tailored for spike camera [6], demonstrating strong adaptability under varying light conditions.

Unlike these empirical or modality-specific approaches, our method is grounded in the physics of event generation. By analytically modeling the relationship between event rate and defocus blur, we reveal the non-monotonic behavior of the event-rate curve and establish a physically interpretable foundation for event-based focusing.

Overall, event-based autofocus represents a promising direction for next-generation adaptive vision systems that require high-speed, low-latency perception and dynamic optical control.

Event Focus Applications. Beyond autofocus, event cameras have shown strong focus-sensitive properties across diverse vision tasks. Lou *et al*. [22] and Teng *et al*. [37] exploited event focal stacks to reconstruct all-in-focus images under complex illumination, while Zhu *et al*. [47] introduced a self-supervised framework that directly restores sharp images from defocused event streams. Jiang *et al*. [17] further leveraged focus variation to infer monocular depth, revealing that defocus cues in event sequences encode rich spatial geometry. At the theoretical level, Haessig *et al*. [14] modeled depth-from-defocus computation using spiking neural networks, highlighting the neuromorphic potential of asynchronous vision.

Although these studies are not designed for explicit autofocus control, they implicitly leverage the relationship between event activity and image sharpness, further confirming the intrinsic connection between event dynamics, depth, and focal clarity.

3. Method

3.1. Principle of Event Cameras and Defocus Behavior

Event cameras asynchronously respond to local brightness changes [5, 20], outputting an event whenever the logarithmic intensity $logI$ variation exceeds a preset contrast threshold C:

$$\Delta L(x, y, t) = L(x, y, t) - L(x, y, t - \delta t) \geq C, \qquad (1)$$

where $L = \log I$ and δt denotes the elapsed time since the last event at pixel (x, y). Each event is represented as (x, y, t, p), with polarity $p \in \{+1, -1\}$ indicating the sign of the intensity change.

During a focus sweep, defocus blur modifies how intensity transitions distribute spatially around edges and textured structures. When the scene is in perfect focus, sharp transitions are spatially compact and only a small set of pixels can cross the temporal threshold [9, 26]. As the lens moves slightly away from focus, blur enlarges the transition zones, allowing more pixels to accumulate sufficient contrast change and trigger events [25]. When defocus becomes severe, local contrast changes weaken and fewer pixels can exceed the threshold [10]. This rise then fall behavior induces a characteristic non-monotonic dependence of event activity on defocus, which we formalize next.

3.2. Event-Rate Modeling and Problem Formulation

As illustrated in Figure 2, we analyze the *event-rate curve* $R(f)$ obtained during a one-way focus sweep and use its *dual-peak–valley* structure to estimate the best focus position $f^\star$. We adopt a locally Gaussian approximation of defocus blur, which captures the essential spatial spreading behavior and is widely used for analytical tractability. We discuss its implications in Appendix A.

Let $\mathcal{E} = \{(x_n, y_n, t_n, p_n)\}_{n=1}^{M}$ denote the event stream and $\{f_i\}_{i=1}^{N}$ the discrete focal positions (or sampled motor steps). Within a temporal window Δt, the raw event rate at position f_i is computed as

$$R_i = \sum_{n=1}^{M} \mathbf{1}\big(|t_n - t(f_i)| \leq \tfrac{\Delta t}{2}\big), \qquad (2)$$

representing the total number of events accumulated around a particular focal distance.

From temporal threshold to a blur-dependent activation region. To connect the temporal triggering rule in Eq. (1) with blur-dependent spatial activation, we adopt a standard short-window approximation during a focus sweep. Let $L_\sigma(\mathbf{x})$ denote the log-intensity image under defocus blur of scale $\sigma \geq 0$, where $\mathbf{x} = (x, y)$. Within a short window Δt,

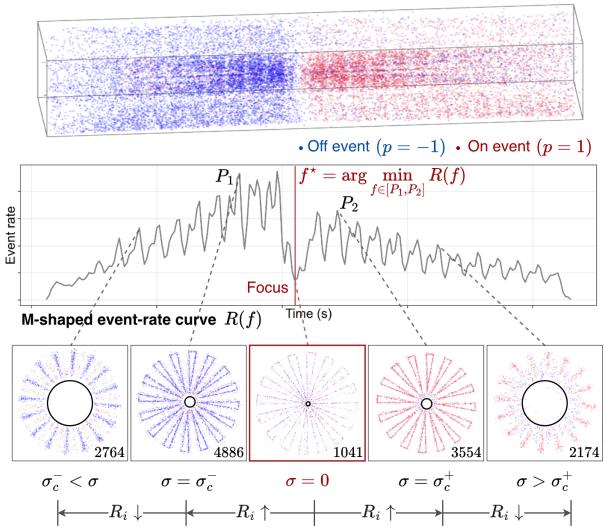


Figure 2. **Structural characterization of the event-rate curve.** Top: spatiotemporal event stream. Middle: during a one-way focus sweep, the event rate $R(f)$ exhibits a dual-peak–valley profile with two dominant peaks (P_1, P_2) and a valley corresponding to the optimal focus $f^\star$. Bottom: accumulated event maps at representative blur levels; smaller central circles denote higher focusing accuracy.

assume that the dominant source of intensity variation is the blur-scale change $\Delta\sigma$ induced by the lens motion. Then, for each pixel,

$$\Delta L(\mathbf{x}; \sigma) \approx \frac{\partial L_\sigma(\mathbf{x})}{\partial \sigma} \Delta\sigma. \qquad (3)$$

Substituting Eq. (3) into the event condition Eq. (1): $|\Delta L| \geq C$ yields

$$\left|\frac{\partial L_\sigma(\mathbf{x})}{\partial \sigma}\right| \geq \theta(C), \qquad \theta(C) := \frac{C}{|\Delta\sigma|}. \qquad (4)$$

The quantity $\theta(C)$ is an *effective* threshold that is constant within the counting window for a fixed sweep speed. It is not a new sensor parameter, but an equivalent form of the same temporal threshold C once the short-window lens-induced blur change $|\Delta\sigma|$ is taken into account.

Define the blur-dependent *activation region*

$$\Omega(\sigma) := \left\{\mathbf{x} : \left|\frac{\partial L_\sigma(\mathbf{x})}{\partial \sigma}\right| \geq \theta(C)\right\}. \qquad (5)$$

Then the event rate in a window is proportional to the number of activated pixels, hence

$$R(\sigma) \propto \mathrm{meas}\big(\Omega(\sigma)\big), \qquad (6)$$

where $\mathrm{meas}(\cdot)$ denotes spatial measure (pixel count in discrete form).

Key theoretical facts. The following facts formalize the rise–peak–fall behavior and its consequence in a one-way sweep.

Proposition 1 (Single-feature rise–peak–fall). For a canonical isolated structure (e.g., a single bright point or a thin line) blurred by a Gaussian kernel, the activation measure $\mathrm{meas}(\Omega(\sigma))$ defined in Eq. (5) is non-monotonic in σ: it increases for small $\sigma > 0$, reaches a maximum at some $\sigma^\star > 0$, and decreases for sufficiently large σ. Moreover, $\sigma = 0$ is a strict local minimum. A complete derivation and closed-form characterization for points and thin lines are provided in Appendix A.

Corollary 1 (Dual-peak–valley in a one-way sweep). In a one-way sweep that crosses the best focus plane, the defocus scale satisfies

$$\sigma(t) \downarrow 0 \uparrow \sigma(t), \tag{7}$$

namely it decreases to $\sigma = 0$ at best focus and increases afterwards. Since $R(\sigma)$ has a local minimum at $\sigma = 0$ and maxima at $\sigma^\star > 0$ on both sides (Proposition 1), the observed event-rate curve $R(f)$ exhibits two dominant peaks (P_1, P_2) separated by a valley, whose minimum corresponds to the sharpest focus $f^\star$.

Problem formulation. Building on Corollary 1, event-based autofocus can be formulated as valley localization within the physically meaningful peak-bounded interval:

$$f^\star = \arg \min_{f \in [P_1, P_2]} R(f). \tag{8}$$

In practice, the measured curve $R(f)$ is often distorted by asynchronous noise, illumination flicker, and mechanical vibration [7, 39], which can obscure the valley structure and lead to inaccurate focus estimation. To address this, we introduce a structural regularization framework that robustly recovers the underlying dual-peak–valley geometry from noisy observations. A controlled simulation under explicitly defined conditions is provided in Appendix B.1, supporting the valley structure under focus sweeping.

3.3. Structural Regularization Framework

Building upon the physical model in Sec. 3.2, we present the **Event Structural Valley-based Autofocus (ESVA)** framework. ESVA regularizes the event-rate curve $R(f)$ through three modules: *structural smoothing*, *consistency filtering*, and *dual-peak constraint*. These operations progressively refine $R(f)$ into a physically consistent dual-peak–valley form from which the optimal focus can be stably derived. The overall ESVA pipeline is summarized in Algorithm 1.

Structural smoothing. The first stage mitigates high-frequency fluctuations caused by the stochastic nature of event generation [10, 13]. A Gaussian kernel is applied to smooth the discrete event-rate sequence:

$$\tilde{R}(f_i) = \frac{\sum_j R(f_j) \exp[-(f_i - f_j)^2/(2\sigma_s^2)]}{\sum_j \exp[-(f_i - f_j)^2/(2\sigma_s^2)]}, \tag{9}$$

where σ_s controls the structural scale. This process suppresses impulsive noise while preserving the global curvature of $R(f)$.

Algorithm 1 Event Structural Valley-based Autofocus (ESVA)

Require: Event stream $\mathcal{E}$ during a focus sweep
Ensure: Estimated focus position $f^\star$ and confidence S
1: Compute raw event-rate curve $R(f)$
2: Apply structural smoothing via Eq. (9)
3: Apply consistency filtering via Eq. (10) and Eq. (11)
4: Detect dominant peaks (P_1, P_2) from local maxima of $\hat{R}(f)$ via Eq. (12) and Eq. (13)
5: Estimate valley $f^\star$ via Eq. (14)
6: Evaluate confidence S via Eq. (15)
7: **Output:** $(f^\star, S)$

Consistency filtering. After smoothing, local consistency across neighboring samples is enforced to maintain physically coherent evolution of the event-rate curve. For each focus step, we compute a normalized deviation metric

$$\delta_i = \frac{|\tilde{R}(f_i) - \tilde{R}(f_{i-1})|}{\max(\tilde{R}(f_{i-1}), \tilde{R}(f_{i+1}))}, \tag{10}$$

which measures the relative jump between consecutive samples. Samples with $\delta_i > \tau_c$ are deemed inconsistent and are projected onto a locally linear manifold:

$$\hat{R}(f_i) = (1 - \eta)\tilde{R}(f_i) + \tfrac{\eta}{2}[\tilde{R}(f_{i-1}) + \tilde{R}(f_{i+1})], \tag{11}$$

where $\eta \in [0, 1]$ controls the filtering strength. This step suppresses isolated oscillations while preserving valid structural transitions in the curve.

Dual-peak constraint. Finally, the feasible focusing interval is restricted to the physically meaningful region bounded by the two dominant peaks of the regularized curve $\hat{R}(f)$. Let $\mathcal{P}$ denote the set of local maxima detected on $\hat{R}(f)$ after structural smoothing and consistency filtering, where standard peak-detection constraints (minimum separation and minimum prominence) are applied.

The primary peak is defined as the maximum in $\mathcal{P}$:

$$P_1 = \arg \max_{f \in \mathcal{P}} \hat{R}(f). \tag{12}$$

The secondary peak P_2 is selected from the remaining candidates in $\mathcal{P} \backslash \{P_1\}$ such that (i) it is sufficiently separated from P_1 along the focus axis, and (ii) its magnitude

is not trivially close to that of P_1. Among candidates satisfying these conditions, the one with the largest response is chosen:

$$P_2 = \arg \max_{f \in \mathcal{P} \setminus \{P_1\}} \hat{R}(f). \tag{13}$$

The resulting search interval is defined as $[P_1, P_2]$ (ordered by their positions along f). This constraint confines valley localization to the peak-bounded region, preventing false minima caused by defocus noise, peak plateaus, or secondary oscillations.

Together, the three modules form a discrete structural regularization process that links the theoretical dual-peak–valley model with its practical focus estimation from real event data.

3.4. Valley Localization and Confidence Evaluation

Within the constrained interval $[P_1, P_2]$, the valley of the regularized event-rate curve $\hat{R}(f)$ corresponds to the physically sharpest focus. The estimated focus is defined as

$$f^\star = \arg \min_{f \in [P_1, P_2]} \hat{R}(f). \tag{14}$$

The associated structural confidence is defined as

$$S = \tfrac{1}{2}\big[\hat{R}(P_1) + \hat{R}(P_2)\big] - \hat{R}(f^\star). \tag{15}$$

The confidence score S quantifies the contrast between the valley and its neighboring peaks, serving as an indicator of focusing stability. A larger S implies stronger structural separation and thus higher focusing reliability. We further validate that S correlates with focusing reliability and can serve as a warning indicator in Appendix B.3.

3.5. Computational Complexity

The proposed ESVA framework scales linearly with the number of focus samples, yielding an overall complexity of $O(N)$. All operations, including smoothing, consistency filtering, and dual-peak detection, are implemented as single-pass procedures without iterative optimization. On standard CPU hardware, the algorithm executes within milliseconds, providing fast and lightweight performance suitable for high-speed event-based autofocus applications.

4. Experiments

4.1. Implementation Details and Comparative Methods

Experimental Parameters. Events are partitioned into fixed temporal windows of $\Delta t = 1\,\text{ms}$ for all datasets. The raw event-rate curve $R(f)$ is Gaussian-smoothed with a kernel width of $\sigma_s = 3$ (in sample units) in structural smoothing process. For consistency filtering, the normalized deviation δ_i is computed, and samples with $\delta_i > \tau_c$ are filtered with $\tau_c = 0.30$ and $\eta = 0.6$. All experiments are conducted on an Intel i9 CPU (3.8 GHz).

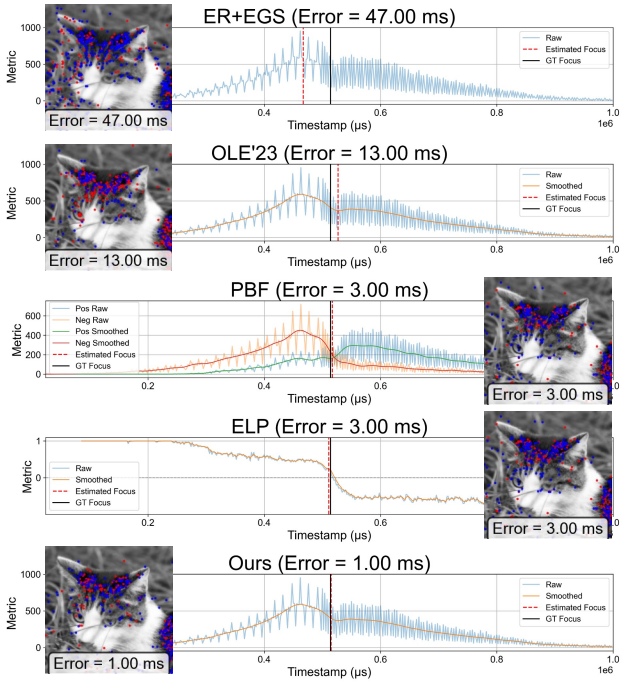


Figure 3. **SYN (cat4-static):** metric curves and estimated focus timestamps. Red dashed: estimate; black solid: ground truth. Subplot titles report timestamp errors; side images visualize events within 1 ms around the estimate.

Table 1. **Results on the SYN dataset.** Average focusing timestamp error (ms) under different motion patterns. Best result per row in **bold**.

Motion Pattern	ER+EGS	OLE'23	PBF	ELP	Ours
Static	42.59	13.66	18.09	8.25	**3.16**
Small Shake	39.59	14.93	11.71	8.36	**5.46**
Huge Shake	26.12	53.14	23.43	**9.43**	11.25
Average	36.10	27.24	17.74	8.68	**6.62**

Comparative methods. We compare with four event-based autofocus methods under identical focal trajectories and event streams: ER+EGS [21], OLE'23 [11], PBF [2], and ELP [3]. Official implementations are used when available (ER+EGS, ELP); others are implemented following the papers. For ELP, we use an event-only variant by detecting sign changes of aggregated positive/negative event counts. We summarize representative formulations and implementation details in Appendix C. Performance is evaluated by **average timestamp error** (ms) on SYN/DAVIS/EVK4 and **average distance error** (μm) on EAD.

4.2. Evaluation on Synthetic Scenarios

Dataset. **SYN** is a public dataset introduced by Bao et al. [3]. It provides simulated focal-sweep sequences

with precisely annotated ground-truth focus timestamps under three motion patterns (*Static*, *Small Shake*, and *Huge Shake*). Each sequence is generated using a physics-based event simulator with controllable blur and motion parameters, offering a clean benchmark for analyzing the structural evolution of event-rate curves.

Results. Figure 3 shows the results on the *cat4-static* scene from the SYN dataset. Our method achieves the smallest temporal deviation and stably identifies a valley near the ground-truth focus. In contrast, the ER+EGS method searches for the maximum event rate and thus tends to select the first local peak before the true focus, resulting in a noticeable offset from the ground-truth position. OLE [11], designed specifically for microscopic imaging, shows limited generalization to complex scenes where noise and motion introduce. Both PBF [2] and ELP [21] are affected by uneven polarity transitions and noise amplification, producing slight deviations from the correct focus. Table 1 reports the quantitative results, where our method achieves an average timestamp error of **6.62 ms**, improving upon the next best method (ELP) by **24%**.

4.3. Evaluation on Real-World Scenarios

We further assess the proposed method on two real event-camera datasets captured using different sensors.

Datasets. **DAVIS** [3] provides synchronized intensity frames and event streams with a spatial resolution of 346×260 pixels, whereas **EVK4** [3] offers higher resolution at 1280×1080 pixels. Both cameras are equipped with motorized varifocal lenses that perform continuous linear focal sweeps. The two datasets differ mainly in spatial resolution and sensor architecture rather than motion speed or scene dynamics, thus providing a consistent foundation for evaluating the proposed method's generalization across different hardware platforms and spatial scales.

DAVIS results. In Figure 4, our method maintains a consistent dual-peak structure with a stable valley that aligns with the ground-truth focus. This indicates that the valley-based representation remains valid in real environments. ER+EGS again stops at an early local peak before reaching the actual focal plane, particularly under illumination changes. OLE'23 is influenced by non-ideal symmetry of event polarity under natural light, resulting in biased estimates. PBF is affected by polarity imbalance due to temporal contrast noise, leading to occasional misalignment of the cross-correlation peak. ELP, though more stable than the others, suffers from small temporal offsets because polarity inversion points drift under flicker or motion. As shown in Table 2, our method achieves an average timestamp error of **1.30 ms**, consistently outperforming all comparisons and maintaining stable convergence across varying illumination and motion conditions.

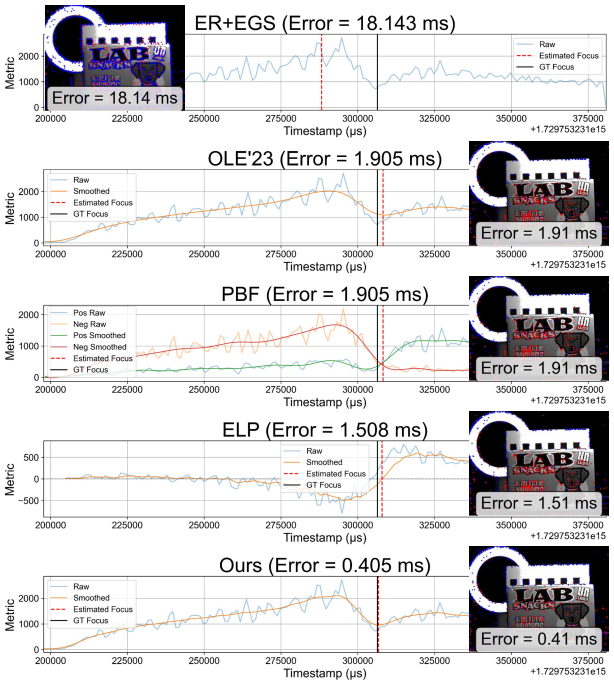


Figure 4. **DAVIS (box-bright-static):** metric curves and estimated focus timestamps. Red dashed: estimate; black solid: ground truth. Side images show events within 1 ms around the estimate.

Table 2. **Results on the DAVIS dataset.** Average focusing timestamp error (ms) across different lighting and motion conditions. Best result per row in **bold**.

Scene Type	ER+EGS	OLE'23	PBF	ELP	Ours
Bright Motion	17.52	14.65	3.51	2.31	**1.36**
Bright Static	18.32	4.76	2.24	1.74	**1.46**
Dark Motion	34.37	7.92	9.47	2.13	**1.01**
Dark Static	34.60	2.71	2.55	1.95	**1.38**
Average	26.20	7.51	4.94	2.04	**1.30**

EVK4 Results. Figure 5 presents the results on the *focusboard-dark-motion* scene from the EVK4 dataset, which features high-resolution recordings under low-light and dynamic conditions. Our method robustly preserves the dual-peak–valley pattern and identifies a stable minimum that aligns with the ground-truth focus, demonstrating reliable performance even under weak illumination fluctuations and motion noise. ER+EGS identifies an incorrect peak due to flattened event-rate curves. OLE'23, constrained by its microscope-oriented polarity symmetry assumption, fails to locate a consistent symmetric in this scene. PBF and ELP

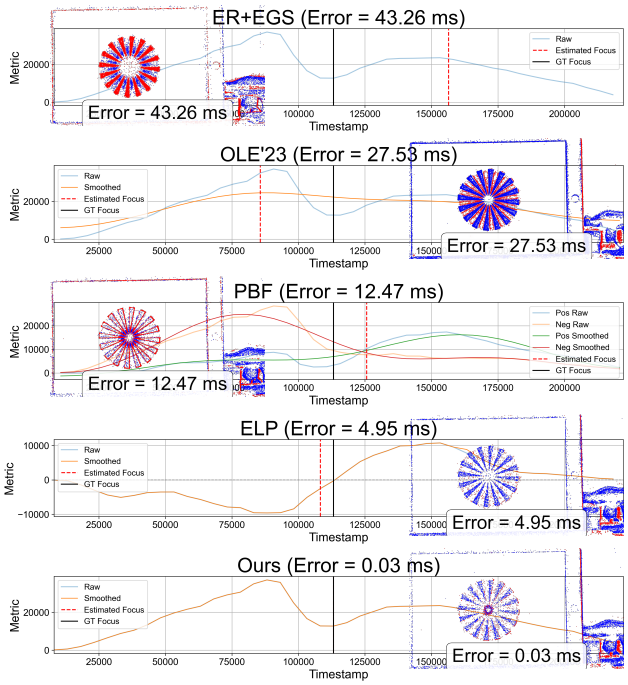


Figure 5. **EVK4 (focusboard-dark-motion):** metric curves and estimated focus timestamps. Red dashed: estimate; black solid: ground truth. Event distributions within 1 ms are visualized around each estimate.

Table 3. **Results on the EVK4 dataset.** Average focusing timestamp error (ms) under various lighting and motion conditions. Best result per row in **bold**.

Scene Type	ER+EGS	OLE'23	PBF	ELP	Ours
Bright Motion	13.96	5.46	4.12	**4.11**	4.34
Bright Static	15.91	3.93	4.79	3.90	**3.01**
Dark Motion	15.64	12.41	13.89	7.85	**4.72**
Dark Static	23.51	17.79	16.17	5.47	**4.80**
Average	17.26	9.90	9.74	5.33	**4.22**

is affected by asynchronous polarity responses, leading to scattered estimates and noticeable deviations from the true focal position. Table 3 reports that our method maintains higher accuracy across all conditions, highlighting its robustness to sensor noise, polarity imbalance, and illumination variation.

4.4. Analysis under Extreme Illumination and Motion Conditions

Finally, we evaluate the method on the most challenging autofocus dataset to highlight the unique advantages of event sensing when conventional images fail.

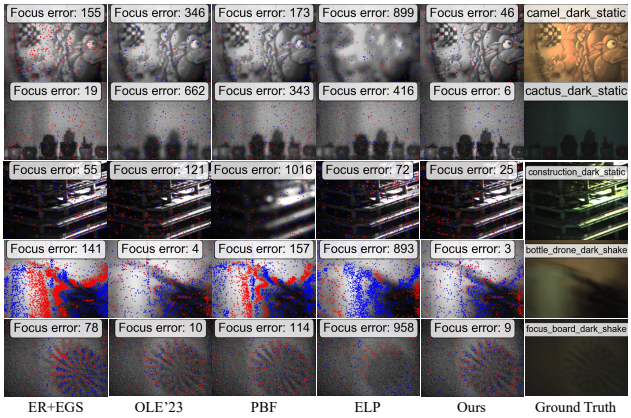


Figure 6. Visualization of **challenging scenarios** from the **EAD dataset**. The top three rows correspond to *dark static* scenes (*camel*, *cactus*, *construction*), and the bottom two rows show more challenging *dark motion* scenes (*bottle drone*, *focusboard*). Each column represents a different autofocus method, with the last column showing the ground-truth image. In such low-light and dynamic conditions, conventional frame-based cameras fail to capture clear images, while event-based methods remain functional. Focus error (in μm) is annotated in each subfigure.

Table 4. **Results on the challenging EAD dataset.** Average focusing distance error (μm) under different illumination and motion conditions. Best result per row in **bold**.

Scene Type	ER+EGS	OLE'23	PBF	ELP	Ours
Light Static	84.57	**64.00**	1232.00	285.65	65.00
Light Shake	87.29	48.71	1029.57	434.93	**46.33**
Dark Static	71.20	329.64	744.36	539.99	**50.00**
Dark Shake	126.14	**53.57**	885.29	642.89	108.29
Average	92.30	123.98	972.81	475.87	**65.38**

Dataset. **EAD** is the event-based autofocus dataset released by Lin et al. [21]. It provides synchronized event streams, image frames, and ground-truth focal positions under four illumination–motion combinations (*light/dark* $\times$ *static/motion*). All sequences are captured using a DAVIS346 sensor with a motorized varifocal lens, covering both normal-light and extremely low-light conditions ($<$ 1 Lux). We use this dataset to evaluate the performance of different methods under realistic and highly challenging settings where frame-based cues become unreliable.

Results. Figure 6, including three dark static scenes (*camel*, *cactus*, *construction*) and two dark motion scenes (*bottle drone*, *focusboard*). Our method consistently yields the most accurate focus across all scenes, maintaining reliable performance even when conventional images are blurred or nearly invisible. Quantitative comparison in Ta-

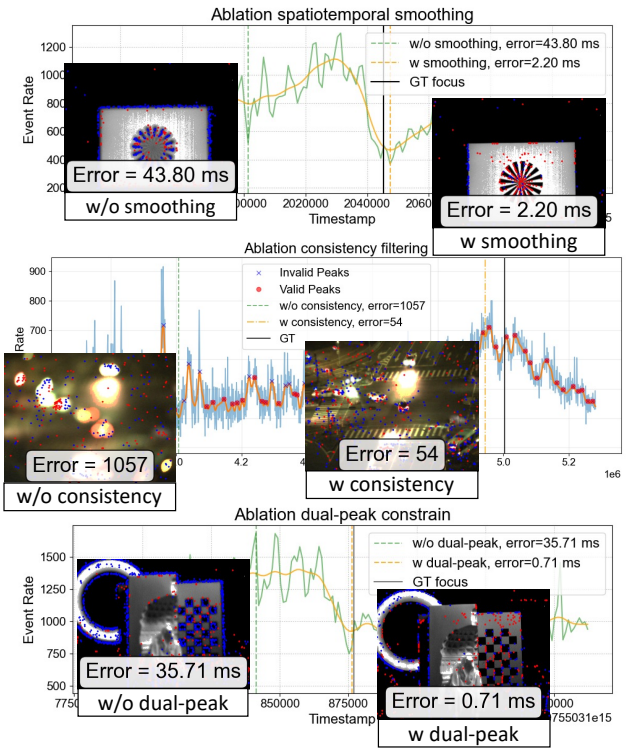


Figure 7. **Ablation study of the proposed structural modules.** (*Top*) **Spatiotemporal smoothing** evaluated on *DAVIS dataset / focusboard-bright-motion* (error in ms). (*Middle*) **Consistency filtering** evaluated on *EAD / crossroad-dark-static* (focus error in μm). (*Bottom*) **Dual-peak constraint** evaluated on *DAVIS dataset / ghost-bright-motion* (error in ms). In each case, the temporal curves compare the metric evolution with and without the corresponding component. The qualitative images show the frame at the estimated focus moment, overlaid with the event distribution (red: positive events; blue: negative events). A clearer frame and sharper event indicate a more accurate focus estimation.

ble 4 shows that our approach achieves the lowest average focusing distance error of **65.38 μm**, improving upon prior methods by nearly **30%**. These results highlight that identifying the focus as the *in-peak minimum* of the event-rate curve, rather than its maximum, enables robust focusing even when frame-based signals are nearly lost.

4.5. Ablation Study

We analyze the contribution of each structural module by selectively disabling it, as shown in Figure 7. Without *spatiotemporal smoothing*, high-frequency oscillations destroy the valley structure, causing large errors ($> 43ms$). Without *consistency filtering*, transient motion noise introduces spurious peaks, increasing the error from $1,057$ to $54\,\mu$m. Removing the *dual-peak constraint* results in false focus at extreme defocus positions. Together, these modules provide complementary stabilization, ensuring an interpretable valley curve and accurate localization.

Table 5. **Runtime comparison (ms) on different event autofucus datasets.**

Dataset	ER+EGS	OLE'23	PBF	ELP	Ours
DAVIS	62.00	1.71	15.67	564.53	**1.43**
EVK4	417.22	2.16	15.95	3699.67	**1.68**

4.6. Runtime Efficiency and Robustness Analysis

Runtime efficiency. All methods are executed purely on CPU without any GPU acceleration to ensure a fair comparison. Table 5 reports the average runtime per sequence across all test scenes on both datasets. The DAVIS dataset contains relatively simple motion and lower spatial resolution, while the EVK4 dataset has a higher resolution and more complex dynamic scenes, resulting in longer processing times for all methods. Compared to the previous event-based methods, our method achieves substantially faster runtime than ER+EGS and ELP, and slightly outperforms OLE'23. This efficiency demonstrates the simplicity and computational advantage of our fully event-domain design.

Additional reliability and tail statistics, failure boundary analysis, and parameter sensitivity experiments are provided in Appendix B.2-B.5.

5. Conclusion and Discussion

We introduce a unified theoretical and practical framework for event-based autofocus. Departing from prior approaches that rely on event-rate maximization or auxiliary image cues, we demonstrate that the physics of event generation naturally give rise to a dual-peak–valley structure in the event-rate curve, whose minimum corresponds to the true focus. Based on this principle, ESVA reliably identifies the physically meaningful valley under diverse motion and illumination conditions, enabling accurate focus estimation without any image reconstruction.

ESVA provides a simple and efficient solution grounded in event-generation physics, using a single structural variable as the focus indicator. While this one-dimensional criterion performs robustly in our evaluations, it can be extended to more complex real-world scenarios by incorporating complementary cues such as polarity patterns, intensity variations, or spatial priors for joint focus estimation. The current formulation assumes a dominant depth layer governs the focusing objective during the sweep. In scenes with multiple competing depth layers, the event-rate curve may exhibit more intricate structures, where additional spatial or task-specific constraints could further improve reliability.

Beyond performance gains, ESVA establishes a physically interpretable foundation for event-driven focusing and may serve as a building block for future neuromorphic vision systems, supporting continuous focus tracking, dynamic zoom control, and high-speed 3D perception.

Acknowledgment

This work was supported by Beijing Science and Technology Plan (Z241100004224011) and the National Natural Science Foundation of China (No. 62425101, No. 62332002). The authors also acknowledge support from the Beijing Key Laboratory of Brain-inspired Spiking Large Models. The authors thank Dr. Yajing Zheng for her valuable suggestions and supportive discussions during the development of this work.

References

[1] Gabriele Balducci, Marco Manzoni, Stefano Tebaldini, Andrea Monti-Guarnieri, Claudio Maria Prati, and Ivan Russo. An improved autofocus algorithm with bayesian tracking of residual motion for automotive mimo-sar imaging. In *Proceedings of the IEEE/CVF International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW)*, pages 1–5. IEEE, 2023. 1

[2] Yuhan Bao, Lei Sun, Yuqin Ma, Diyang Gu, and Kaiwei Wang. Improving fast auto-focus with event polarity. *Optics Express*, 31(15):24025–24044, 2023. 2, 5, 6

[3] Yuhan Bao, Shaohua Gao, Wenyong Li, and Kaiwei Wang. One-step event-driven high-speed autofocus. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6222–6230, 2025. 5, 6

[4] Patrick Bardow, Andrew J Davison, and Stefan Leutenegger. Simultaneous optical flow and intensity estimation from an event camera. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 884–892, 2016.

[5] Christian Brandli, Raphael Berner, Minhao Yang, Shih-Chii Liu, and Tobi Delbruck. A 240× 180 130 db 3 μs latency global shutter spatiotemporal vision sensor. *IEEE Journal of Solid-State Circuits*, 49(10):2333–2341, 2014. 1, 3

[6] Siwei Dong, Lin Zhu, Daoyuan Xu, Yonghong Tian, and Tiejun Huang. An efficient coding method for spike camera using inter-spike intervals. In *Data Compression Conference (DCC)*, pages 568–568, 2019. 2

[7] Peiqi Duan, Zihao W. Wang, Xinyu Zhou, Yi Ma, and Boxin Shi. Eventzoom: Learning to denoise and super resolve neuromorphic events. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12819–12828, 2021. 4

[8] Lawrence Firestone, Kitty Cook, Kevin Culp, Neil Talsania, and Kendall Preston Jr. Comparison of autofocus methods for automated microscopy. *Cytometry: The Journal of the International Society for Analytical Cytology*, 12(3):195–206, 1991. 1

[9] Guillermo Gallego, Henri Rebecq, and Davide Scaramuzza. A unifying contrast maximization framework for event cameras, with applications to motion, depth, and optical flow estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3867–3876, 2018. 1, 2, 3

[10] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J. Davison, Jörg Conradt, Kostas Daniilidis, and Davide Scaramuzza. Event-based vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(1):154–180, 2020. 1, 3, 4

[11] Zhou Ge, Haoyu Wei, Feng Xu, Yizhao Gao, Zhiqin Chu, Hayden K.-H. So, and Edmund Y. Lam. Millisecond autofocusing microscopy using neuromorphic event sensing. *Optics and Lasers in Engineering*, 160:107247, 2023. 2, 5, 6

[12] Henry A Greene, Robert Beadles, and Jaroslav Pekar. Challenges in applying autofocus technology to low vision telescopes. *Optometry and Vision Science*, 69(1):25–31, 1992. 1

[13] Shasha Guo and Tobi Delbruck. Low cost and latency event camera background activity denoising. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(1):785–795, 2023. 4

[14] Guillaume Haessig, Xavier Berthelon, Sio-Hoi Ieng, and Ryad Benosman. A spiking neural network model of depth from defocus for event-based neuromorphic vision. *Scientific Reports*, 9(1):3744, 2019. 2

[15] Charles Herrmann, Richard Strong Bowen, Neal Wadhwa, Rahul Garg, Qiurui He, Jonathan T. Barron, and Ramin Zabih. Learning to autofocus. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 1

[16] Wei-Yao Hsu, Chien-Shing Lee, Po-Jui Chen, Nien-Tsu Chen, Fong-Zhi Chen, Zong-Ru Yu, Ching-Hsiang Kuo, and Chi-Hung Hwang. Development of the fast astigmatic autofocus microscope system. *Measurement Science and Technology*, 20(4):045902, 2009. 1

[17] Chenxu Jiang, Mingyuan Lin, Chi Zhang, Zhenghai Wang, and Lei Yu. Learning monocular depth from focus with event focal stack. *arXiv preprint arXiv:2405.06944*, 2024. 2

[18] Chenxu Jiang, Mingyuan Lin, Chi Zhang, Zhenghai Wang, and Lei Yu. Learning monocular depth from focus with event focal stack, 2024.

[19] Pritam P Karmokar, Quan H Nguyen, and William J Beksi. Secrets of edge-informed contrast maximization for event-based vision. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 630–639. IEEE, 2025. 2

[20] Patrick Lichtsteiner, Christoph Posch, and T. Delbrück. A 128×128 120 db 15 μs latency asynchronous temporal contrast vision sensor. *IEEE Journal of Solid-State Circuits*, 43 (2):566–576, 2008. 1, 2, 3

[21] Shijie Lin, Yinqiang Zhang, Lei Yu, Bin Zhou, Xiaowei Luo, and Jia Pan. Autofocus for event cameras. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16344–16353, 2022. 1, 2, 5, 6, 7

[22] Hanyue Lou, Minggui Teng, Yixin Yang, and Boxin Shi. All-in-focus imaging from event focal stack. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17366–17375, 2023. 1, 2

[23] Yunfei Luan, Yating Luo, Yuxuan Liu, Musen Zhang, Yujian An, Jianxin Yang, Yao Guo, and Guang-Zhong Yang. Autofocusing with 3-d tracking for robot-assisted microsurgery. *IEEE/ASME Transactions on Mechatronics*, 2025. 1

[24] Florian Mahlknecht, Daniel Gehrig, Jeremy Nash, Friedrich M. Rockenbauer, Benjamin Morrell, Jeff Delaune, and Davide Scaramuzza. Exploring event camera-based odometry for planetary robots. *IEEE Robotics and Automation Letters*, 7(4):8651–8658, 2022. 1

[25] Liyuan Pan, Cedric Scheerlinck, Xin Yu, Richard Hartley, Miaomiao Liu, and Yuchao Dai. Bringing a blurry frame alive at high frame-rate with an event camera. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6820–6829, 2019. 1, 3

[26] Xin Peng, Ling Gao, Yifu Wang, and Laurent Kneip. Globally-optimal contrast maximisation for event cameras. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(7):3479–3495, 2022. 3

[27] Said Pertuz, Domenec Puig, and Miguel Angel Garcia. Analysis of focus measure operators for shape-from-focus. *Pattern Recognition*, 46(5):1415–1432, 2013.

[28] Henry Pinkard, Zachary Phillips, Arman Babakhani, Daniel A Fletcher, and Laura Waller. Deep learning for single-shot autofocus microscopy. *Optica*, 6(6):794–797, 2019. 1

[29] Xing Qu, Chunyang Ma, Wenbo Hu, Huiwen Den, and Shuming Yang. A robust autofocusing method for microscopic imaging based on an event camera. *Optics and Lasers in Engineering*, 175:108025, 2024. 2

[30] Henri Rebecq, Timo Horstschäfer, Guillermo Gallego, and Davide Scaramuzza. Evo: A geometric approach to event-based 6-dof parallel tracking and mapping in real time. *IEEE Robotics and Automation Letters*, 2(2):593–600, 2016.

[31] Henri Rebecq, Timo Horstschaefer, Guillermo Gallego, and Davide Scaramuzza. Evo: A geometric approach to event-based 6-dof parallel tracking and mapping in real time. *IEEE Robotics and Automation Letters*, 2(2):593–600, 2017. 1

[32] Henri Rebecq, Guillermo Gallego, Elias Mueggler, and Davide Scaramuzza. Emvs: Event-based multi-view stereo—3d reconstruction with an event camera in real-time. *International Journal of Computer Vision*, 126(12):1394–1414, 2018. 1

[33] Henri Rebecq, René Ranftl, Vladlen Koltun, and Davide Scaramuzza. High speed and high dynamic range video with an event camera. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(6):1964–1980, 2021. 1

[34] Murat Selek. A new autofocusing method based on brightness and contrast for color cameras. *Advances in Electrical and Computer Engineering*, 16:39–44, 2016.

[35] Shintaro Shiba, Yannick Klose, Yoshimitsu Aoki, and Guillermo Gallego. Secrets of event-based optical flow, depth and ego-motion estimation by contrast maximization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):7742–7759, 2024. 1

[36] Changqing Su, Zhiyuan Ye, Yongsheng Xiao, You Zhou, Zhen Cheng, Bo Xiong, Zhaofei Yu, and Tiejun Huang. Intensity-robust autofocus for spike camera. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 25018–25027, 2024. 2

[37] Mingjie Teng, Hao Lou, Yuxuan Yang, Tiejun Huang, and Boxin Shi. Hybrid all-in-focus imaging from neuromorphic focal stack. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):10124–10137, 2024. 2

[38] Chengyu Wang, Qian Huang, Ming Cheng, Zhan Ma, and David J Brady. Deep learning for camera autofocus. *IEEE Transactions on Computational Imaging*, 7:258–271, 2021. 1

[39] Yanxiang Wang, Bowen Du, Yiran Shen, Kai Wu, Guangrong Zhao, Jianguo Sun, and Hongkai Wen. Ev-gait: Event-based robust gait recognition using dynamic vision sensors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6351–6360, 2019. 4

[40] Xijie Xiang, Lin Zhu, Jianing Li, Yixuan Wang, Tiejun Huang, and Yonghong Tian. Learning super-resolution reconstruction for high temporal resolution spike stream. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(1):16–29, 2023. 1

[41] Chenxiao Yin, Kun Deng, Lu Zhang, Xinran Tian, Zhanye Chen, and Yan Huang. A minimum rényi entropy autofocus for automotive sar imaging. In *Proceedings of the IEEE MTT-S International Wireless Symposium (IWS)*, pages 1–3. IEEE, 2025. 1

[42] Tianyi Zhang, Kohei Shimasaki, Idaku Ishii, and Akio Namiki. Dynamic-range focal sweep: Seamless continuous autofocus based on high-speed vision for magnified object tracking. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4838–4845. IEEE, 2024. 1

[43] Yupeng Zhang, Liyan Liu, Weitao Gong, Haihua Yu, Wei Wang, Chongying Zhao, Peng Wang, and Toshitsugu Ueda. Autofocus system and evaluation methodologies: A literature review. *Sensors & Materials*, 30, 2018. 1

[44] Zheng Zhang, Yu Liu, Zhihui Xiong, Jing Li, and Maojun Zhang. Focus and blurriness measure using reorganized dct coefficients for an autofocus application. *IEEE Transactions on Circuits and Systems for Video Technology*, 28(1):15–30, 2016. 1

[45] Yi Zhou, Guillermo Gallego, Henri Rebecq, Lukas Kneip, Hongdong Li, and Davide Scaramuzza. Semi-dense 3d reconstruction with a stereo event camera. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 235–251, 2018. 1

[46] Lin Zhu, Xiao Wang, Yitian Chang, Jinxiu Li, Tiejun Huang, and Yonghong Tian. Event-based video reconstruction via potential-assisted spiking neural network. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3594–3604, 2022. 1

[47] Lin Zhu, Xiantao Ma, Xiao Wang, Lizhi Wang, and Hua Huang. Evfocus: Learning to reconstruct sharp images from out-of-focus event streams. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. 2